

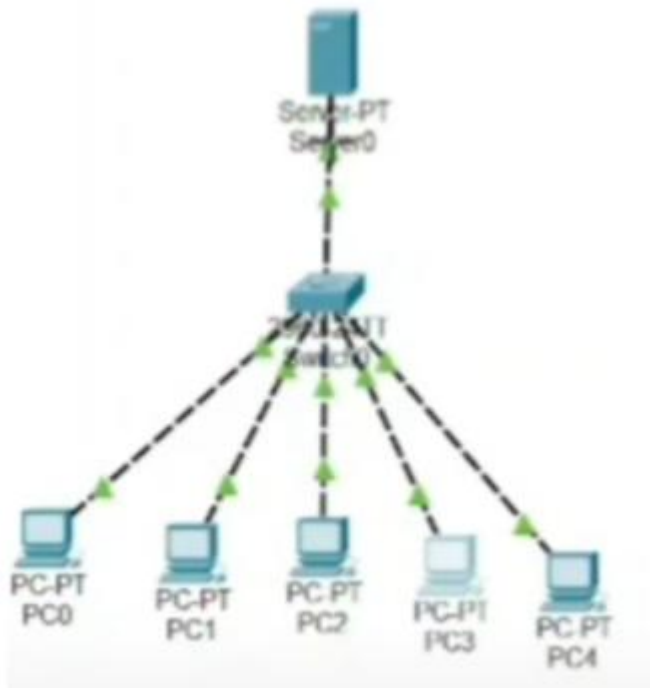
Computer Networks: Assignment-7

Name: Suraj Shah

Admission Number: I24AI012

Q 1) Construct the below network in the Cisco packet tracer and connect the server and Pcs using DHCP protocol.

Diagram:



IP Config:

Server0: 192.168.1.1 / 255.255.255.0

DHCP Service:

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
LAN_POOL	192.168.1.1	192.168.1.1	192.168.1.10	255.255.2...	50	0.0.0.0	0.0.0.0

Received PC IPs:

PC0: 192.168.1.5 / 255.255.255.0

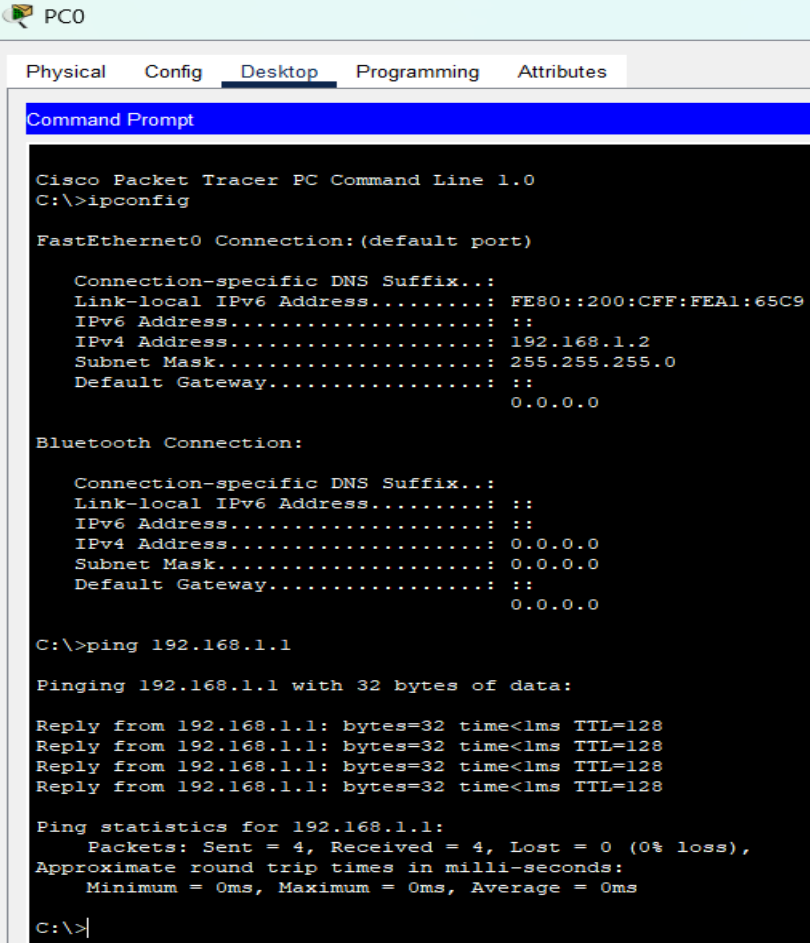
PC1: 192.168.1.2 / 255.255.255.0

PC2: 192.168.1.4 / 255.255.255.0

PC3: 192.168.1.6 / 255.255.255.0

PC4: 192.168.1.3 / 255.255.255.0

Output:



```
PC0
Physical  Config  Desktop  Programming  Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address.....: FE80::200:CFF:FEA1:65C9
    IPv6 Address.....: ::
    IPv4 Address.....: 192.168.1.2
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                                0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address.....: ::
    IPv6 Address.....: ::
    IPv4 Address.....: 0.0.0.0
    Subnet Mask.....: 0.0.0.0
    Default Gateway.....: ::
                                0.0.0.0

C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

C:\>ping 192.168.1.3

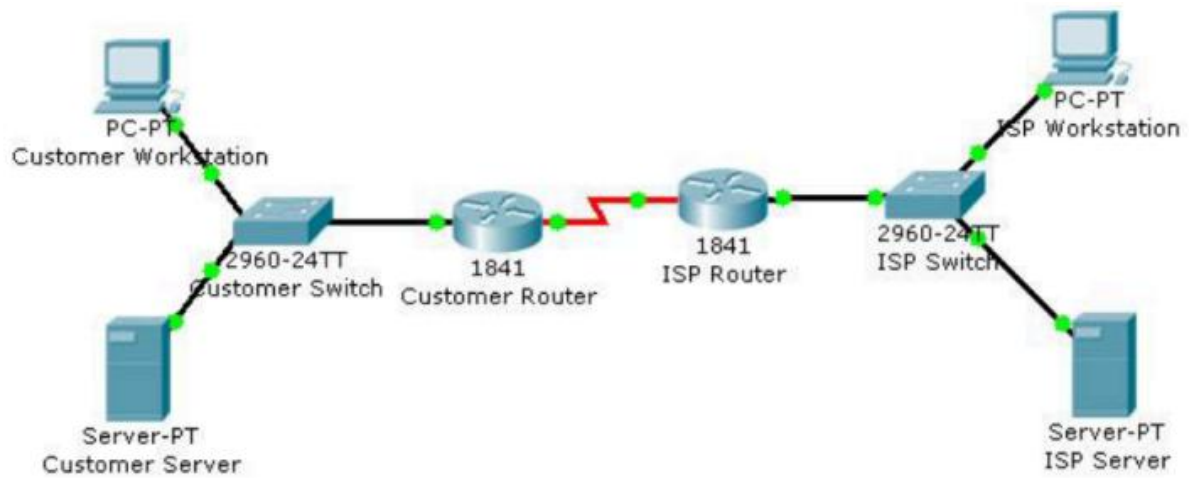
Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Q 2) Construct the below network in the Cisco packet tracer and Configuring a Cisco Router as a DHCP Server

Diagram:



IP Config:

Customer Router:

FastEthernet0/0: 192.168.1.1 / 255.255.255.0

Serial0/1/0: 10.0.0.1/ 255.255.255.252

ISP Router:

FastEthernet0/0: 192.168.2.1 / 255.255.255.0

Serial0/1/0: 10.0.0.2/ 255.255.255.252

ISP Workstation: 192.168.2.2 / 255.255.255.0

Commands:

Customer Router:

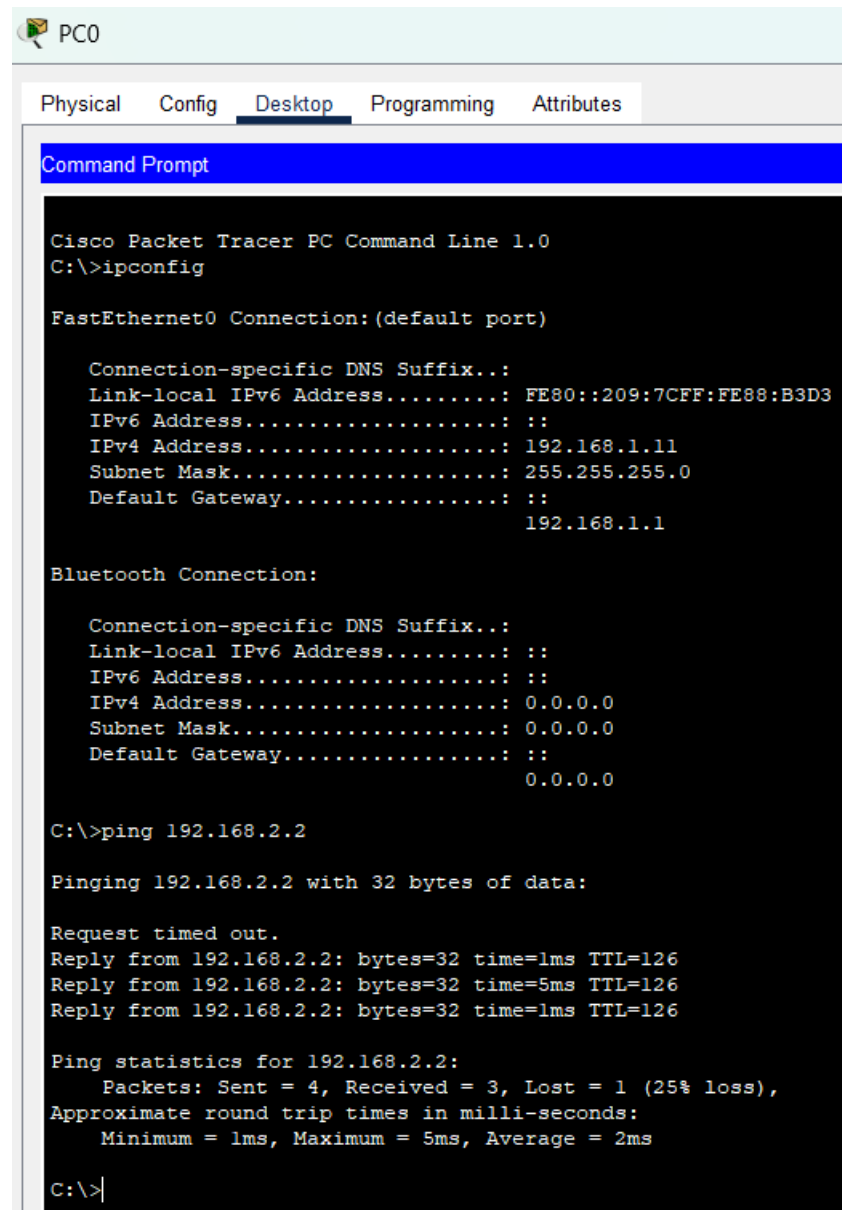
```
ip dhcp excluded-address 192.168.1.1 192.168.1.10
```

```
ip dhcp pool CUSTOMER_LAN
network 192.168.1.0 255.255.255.0
default-router 192.168.1.1
dns-server 8.8.8.8
exit
```

Received PC IP:

Customer Workstation: 192.168.1.11 / 255.255.255.0

Output:



The screenshot shows the 'PC0' configuration window in Cisco Packet Tracer. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The text in the Command Prompt is as follows:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::209:7CFF:FE88:B3D3
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.1.11
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::
                                   192.168.1.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=5ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 5ms, Average = 2ms

C:\>|
```